

Theory section (test build)

1 Theory of planning with stochastic latent world models

This section collects the formal statements behind the empirical findings. Every result is labelled *Known* (standard, cited, proof recalled only when it is short) or *New* (elementary, proved here; “new” means new in this form for the planning setting, not deep). All formulas were checked numerically with `experiments/d4_verify.py` (Section 1.7). Where the numerics forced a weaker statement than first conjectured, this is said explicitly (Remark 27).

Result	Where	One-line statement	Status
T1	Thm. 5	regret of deterministic (mean-embedding) planning = difference of noise premiums, $\leq$ their spread, tight	New
	Cor. 7, 8	failure-penalised cost (regret linear in κ); CVaR $_{\alpha}$ (planner blind to α)	New
T2	Thm. 10	$\mathbb{P}(\text{risky selected}) = 1 - (1 - F_{\text{Bin}}(k^*; M, p))^{N_r}$, monotone in N_r , $\rightarrow 1$	New
	Thm. 11	excess true cost $\kappa(p - \theta)\mathbb{P}(\text{risky})$; optimism $\rightarrow \kappa p$	New
	Thm. 13	fresh-sample verification: risk $1 - (1 - F')^K$, free of N_r	New
	Prop. 15	monotone shrinkage with equal M cannot change the argmin; unequal M can	New
T3	Thm. 17	energy score strictly proper	Known
	Prop. 18	law identified; noise map and cross-candidate coupling are not	New
	Thm. 19	set-consistency of the empirical energy-score minimiser	New (standard proof)
T4	Prop. 22, Thm. 24	variance $t\sigma^2$ vs bounded; open-loop value = $H \times$ closed-loop value	New
	Thm. 26	strict open-loop gap with an absorbing boundary	New
T5	Thm. 28	no scale recalibration repairs a change of shape; explicit floors	New

Table 1: Summary of results and provenance.

1.1 Setting and notation

The latent state is $z \in \mathbb{R}^d$. A (stochastic) world model predicts $z' = f(z, a, u)$ with driving noise $u \sim \gamma := \mathcal{N}(0, I_k)$ drawn independently at every step; a deterministic model is the special case without u . Fix the current latent state s and a candidate set $\mathcal{A}$ of action sequences ($|\mathcal{A}| = N$; everything below holds for any $\mathcal{A}$ on which the stated minima are attained). For $a \in \mathcal{A}$ let $Z_a = Z_H(s, a, U)$ be the terminal latent with law P_a (finite second moment), mean μ_a and covariance Σ_a . Let $g \in \mathbb{R}^d$ be the goal, $c(z) = \|z - g\|^2$, let $F \subset \mathbb{R}^d$ be a measurable failure set, $\kappa \geq 0$, and

$$c_{\kappa}(z) = \|z - g\|^2 + \kappa \mathbf{1}_F(z), \quad p_a = \mathbb{P}(Z_a \in F).$$

For $\alpha \in [0, 1)$ and an integrable random variable Y , $\text{CVaR}_{\alpha}(Y) = \min_{t \in \mathbb{R}} \{t + (1 - \alpha)^{-1} \mathbb{E}(Y - t)_+\}$ [10, 11]; it equals $\mathbb{E}[Y \mid Y \geq \text{VaR}_{\alpha}(Y)]$ if Y has a continuous law, equals $\mathbb{E}Y$ for $\alpha = 0$, equals y

for a constant $Y \equiv y$, and satisfies $\text{CVaR}_\alpha(Y) \geq \mathbb{E}Y$. The planning criterion is a functional ρ of the law P_a :

$$\rho_{\mathbb{E}}(P_a) = \mathbb{E} c_\kappa(Z_a) =: J_\kappa(a), \quad \rho_\alpha(P_a) = \text{CVaR}_\alpha(c_\kappa(Z_a)).$$

A *particle planner* draws M i.i.d. copies $Z_a^{(1)}, \dots, Z_a^{(M)}$ and scores $\hat{J}(a) = M^{-1} \sum_j c_\kappa(Z_a^{(j)})$; a *deterministic (mean-embedding) planner* uses a predictor $m(a) \in \mathbb{R}^d$ and scores the single point $c_\kappa(m(a))$. Since $\rho_{\mathbb{E}}(\delta_x) = \rho_\alpha(\delta_x) = c_\kappa(x)$ for a point mass, *the deterministic planner does not depend on which of these criteria is intended*; this is the root of the results of Section 1.2. Ties in an $\arg \min$ are resolved arbitrarily unless stated.

1.2 T1: the variance gap of mean-embedding planning

Proposition 1 (Bias–variance decomposition; Known). *If $\mathbb{E}\|Z\|^2 < \infty$ with mean μ and covariance Σ , then $\mathbb{E}\|Z - g\|^2 = \|\mu - g\|^2 + \text{tr } \Sigma$.*

Proof. Write $Z - g = (Z - \mu) + (\mu - g)$; the cross term has mean $2\mathbb{E}\langle Z - \mu, \mu - g \rangle = 0$ and $\mathbb{E}\|Z - \mu\|^2 = \text{tr } \Sigma$. $\square$

Proposition 2 (MSE-trained predictors converge to the conditional mean; Known). *Let (X, Y) be random with $\mathbb{E}\|Y\|^2 < \infty$, $X = (s, a)$, $\mu(X) = \mathbb{E}[Y \mid X]$ and $\Sigma(X) = \text{Cov}(Y \mid X)$. For every measurable m with $\mathbb{E}\|m(X)\|^2 < \infty$,*

$$L(m) := \mathbb{E}\|Y - m(X)\|^2 = L(\mu) + \mathbb{E}\|m(X) - \mu(X)\|^2, \quad L(\mu) = \mathbb{E} \text{tr } \Sigma(X).$$

Hence μ is the a.s. unique minimiser, and if $L(m_n) \rightarrow L(\mu)$ along a sequence of predictors then $m_n \rightarrow \mu$ in $L^2(P_X)$. The residual loss of the best deterministic predictor equals the average noise trace.

Proof. Condition on X and apply Proposition 1 with $g = m(X)$: $\mathbb{E}[\|Y - m(X)\|^2 \mid X] = \|\mu(X) - m(X)\|^2 + \text{tr } \Sigma(X)$; take expectations. $\square$

Remark 3 (Multi-step rollouts). *Proposition 2 applies to the H -step predictor $s, a \mapsto \mathbb{E}[Z_H \mid s, a]$. A one-step MSE predictor $\bar{f}(z, a) = \mathbb{E}[f(z, a, U)]$ composed H times equals $\mathbb{E}Z_H$ only if the dynamics are affine in z (Jensen’s inequality otherwise), so the rollout of a deterministic model carries an additional bias. The theorem below is stated for an arbitrary predictor m so that this bias is covered.*

Definition 4 (Noise premium). *For a criterion ρ and predictor m put $d(a) = c_\kappa(m(a))$ (the deterministic score) and*

$$\tau_\rho(a) = \rho(P_a) - d(a) \quad (\text{“noise premium” of } a).$$

For $\rho_{\mathbb{E}}$, $\kappa = 0$ and $m = \mu$ one has $\tau_{\mathbb{E}}(a) = \text{tr } \Sigma_a =: v_a$ by Proposition 1. Write $\text{osc}_{\mathcal{A}} \tau = \max_a \tau(a) - \min_a \tau(a)$ and $\Delta_v = \text{osc}_{\mathcal{A}} v$ (the variance spread).

Theorem 5 (Regret of mean-embedding planning; New). *Let $a_d \in \arg \min_a d(a)$ and $a^* \in \arg \min_a \rho(P_a)$, and let $R = \rho(P_{a_d}) - \rho(P_{a^*})$ be the regret of the deterministic planner under criterion ρ .*

- (a) (exact) $R = [d(a_d) - d(a^*)] + [\tau_\rho(a_d) - \tau_\rho(a^*)]$, and the first bracket is ≤ 0 .
- (b) (bound) $0 \leq R \leq \tau_\rho(a_d) - \tau_\rho(a^*) \leq \text{osc}_{\mathcal{A}} \tau_\rho$. In particular $R = 0$ if τ_ρ is constant on $\mathcal{A}$ (homoscedastic noise).
- (c) (tightness) For every $\Delta > 0$ and $\eta \in (0, \Delta)$ there is a two-candidate instance with $\rho = \rho_{\mathbb{E}}$, $\kappa = 0$, $m = \mu$, $\text{osc } \tau = \Delta_v = \Delta$ and $R = \Delta - \eta$. Hence the constant 1 in $R \leq \Delta_v$ cannot be improved, and the relative regret $R/\rho(P_{a^*}) = (\Delta - \eta)/\eta$ is unbounded as $\eta \rightarrow 0$.

Proof. (a),(b): by definition $\rho(P_a) = d(a) + \tau_\rho(a)$ for every a , so $R = d(a_d) - d(a^*) + \tau_\rho(a_d) - \tau_\rho(a^*)$, and $d(a_d) \leq d(a^*)$ by the choice of a_d . $R \geq 0$ because a^* minimises $\rho(P_a)$. The last inequality is the definition of oscillation. (c): Take $d = 1$, $g = 0$. Candidate 1: $Z_1 = \sqrt{\Delta}\xi$, $\xi \sim \mathcal{N}(0, 1)$ (so $\mu_1 = 0$, $v_1 = \Delta$). Candidate 2: $Z_2 \equiv \sqrt{\eta}$ (so $v_2 = 0$). With $m = \mu$ the deterministic scores are $0 < \eta$, hence $a_d = 1$, but $J(1) = \Delta > \eta = J(2)$, so $a^* = 2$ and $R = \Delta - \eta$, $\text{osc } v = \Delta$. $\square$

Corollary 6 (Expected cost, with predictor error; New). *Let $\kappa = 0$, $\rho = \rho_E$ and suppose $\sup_a |c(m(a)) - c(\mu_a)| \leq \beta$. Then $R \leq \Delta_v + 2\beta$; for $m = \mu$ (the infinite-data limit of MSE training, Prop. 2), $R \leq v_{a_d} - v_{a^*} \leq \Delta_v$.*

Proof. $\tau_E(a) = v_a + [c(\mu_a) - c(m(a))]$ and the bracket lies in $[-\beta, \beta]$, so $\text{osc } \tau_E \leq \Delta_v + 2\beta$; apply Theorem 5(b). $\square$

Corollary 7 (Failure-penalised cost; New). *Let $\rho = \rho_E$, $\kappa \geq 0$, $m = \mu$. Then $\tau_\kappa(a) = v_a + \kappa(p_a - \mathbf{1}_F(\mu_a))$ and $R \leq \tau_\kappa(a_d) - \tau_\kappa(a^*)$. If no mean lies in F (the typical situation: the mean-embedding never “sees” the hazard), then $R \leq \text{osc}_\mathcal{A}(v + \kappa p)$. This is tight, and the regret can grow linearly in κ : in $d = 1$ with $F = [B, \infty)$, $g < B$, let $Z_1 \sim \mathcal{N}(g, s^2)$ and $Z_2 \equiv g - \varepsilon$ with $\varepsilon > 0$. The deterministic planner picks candidate 1 (score 0 versus ε^2) and, whenever $s^2 + \kappa p_1 \geq \varepsilon^2$ (so that candidate 2 is the true optimum),*

$$R(\kappa) = s^2 + \kappa \bar{\Phi}\left(\frac{B - g}{s}\right) - \varepsilon^2 \xrightarrow{\kappa \rightarrow \infty} \infty.$$

Proof. $\mathbb{E}c_\kappa(Z_a) = \|\mu_a - g\|^2 + v_a + \kappa p_a$ and $d(a) = \|\mu_a - g\|^2 + \kappa \mathbf{1}_F(\mu_a)$; subtract and use Theorem 5. In the example $J(1) = s^2 + \kappa \mathbb{P}(Z_1 \geq B)$, $J(2) = \varepsilon^2$. $\square$

Corollary 8 (CVaR $_\alpha$; New). *Let $\kappa = 0$, $\rho = \rho_\alpha$, $m = \mu$. The deterministic planner is the same for every α , its premium is $\tau_\alpha(a) = \text{CVaR}_\alpha(c(Z_a)) - \|\mu_a - g\|^2 \geq v_a$, and $R_\alpha \leq \text{osc}_\mathcal{A} \tau_\alpha$. In the one-dimensional Gaussian example ($Z_1 \sim \mathcal{N}(g, s^2)$, $Z_2 \equiv g + \varepsilon$, $0 < \varepsilon^2 < s^2 c_\alpha$)*

$$R_\alpha = s^2 c_\alpha - \varepsilon^2, \quad c_\alpha = 1 + \frac{z_\alpha \varphi(z_\alpha)}{\bar{\Phi}(z_\alpha)} \geq 1 + z_\alpha^2, \quad z_\alpha = \Phi^{-1}\left(\frac{1+\alpha}{2}\right),$$

where φ, Φ are the standard normal density and distribution function. Thus c_α increases with α and $c_\alpha \sim 2 \ln \frac{1}{1-\alpha}$ as $\alpha \rightarrow 1$; the CVaR-regret of the deterministic planner exceeds its expected-cost regret $s^2 - \varepsilon^2$ by the factor $\approx c_\alpha$.

Proof. $\text{CVaR}_\alpha(Y) \geq \mathbb{E}Y$ gives $\tau_\alpha \geq v$. For the example, $c(Z_1) = s^2 \xi^2$ with $\xi \sim \mathcal{N}(0, 1)$, so $\text{CVaR}_\alpha(c(Z_1)) = s^2 \text{CVaR}_\alpha(\xi^2)$. ξ^2 has a continuous law with $\mathbb{P}(\xi^2 > z^2) = 2\bar{\Phi}(z) = 1 - \alpha$ exactly when $z = z_\alpha$, hence $\text{CVaR}_\alpha(\xi^2) = \frac{1}{1-\alpha} \mathbb{E}[\xi^2; |\xi| > z_\alpha] = \frac{2}{1-\alpha} \int_{z_\alpha}^\infty x^2 \varphi(x) dx = \frac{2[z_\alpha \varphi(z_\alpha) + \bar{\Phi}(z_\alpha)]}{2\bar{\Phi}(z_\alpha)} = c_\alpha$, using $\int_z^\infty x^2 \varphi(x) dx = z\varphi(z) + \bar{\Phi}(z)$. The inequality $c_\alpha \geq 1 + z_\alpha^2$ is Mills' bound $\bar{\Phi}(z) \leq \varphi(z)/z$. Candidate 2 is deterministic with cost ε^2 for all α ; the deterministic planner picks candidate 1 (score 0). $\square$

Remark 9. *Theorem 5 says that the price of ignoring the predictive spread is controlled by how much the noise premium varies across the candidates being compared. This is why a bimodal-wind cliff (spread depends on the distance to the pit, p_a varies strongly) punishes the mean-embedding planner, whereas a model with homoscedastic additive noise would not. The regret is a statement about the objective, not about estimation: even a perfect mean predictor incurs it.*

1.3 T2: tail blindness amplified by selection

Model. There are $N_r \geq 1$ *risky* candidates and one *safe* candidate. A particle of a risky candidate costs $c_r + \kappa\xi$, with $\xi \sim \text{Bernoulli}(p)$, $0 < p < 1$, all particles of all candidates independent (no common random numbers); the safe candidate costs $c_s > c_r$ with certainty. Put

$$\theta = \frac{c_s - c_r}{\kappa}, \quad k^* = \lfloor M\theta \rfloor, \quad J_r = c_r + \kappa p,$$

so that $J_r > c_s \iff p > \theta$ (the *mistake regime*: the risky candidates are truly worse). With M particles a risky candidate has estimate $\hat{J}_i = c_r + \kappa K_i/M$, $K_i \sim \text{Bin}(M, p)$ i.i.d. Let $F := F_{\text{Bin}}(k^*; M, p) = \mathbb{P}(K \leq k^*)$. The planner selects the minimum estimate; a tie between the safe candidate and a risky one is resolved in favour of the risky one (the opposite convention replaces k^* by $\lceil M\theta \rceil - 1$ everywhere).

Theorem 10 (Selection probability; New). (a) $\mathbb{P}(\text{risky selected}) = 1 - (1 - F)^{N_r}$.

(b) This is nondecreasing in N_r and strictly increasing when $0 < F < 1$ (equivalently $\theta < 1$); $F \geq (1 - p)^M > 0$ with equality iff $k^* = 0$.

(c) $1 - \mathbb{P}(\text{risky selected}) = \exp(-\lambda N_r)$ with $\lambda = -\ln(1 - F) \geq F$; the smallest N_r with $\mathbb{P}(\text{risky selected}) \geq 1 - \delta$ is $\lceil \ln(1/\delta)/\lambda \rceil \approx \ln(1/\delta)/F$. In particular $\mathbb{P} \rightarrow 1$ geometrically as $N_r \rightarrow \infty$ for every fixed M .

(d) (particle budget) If $\theta < p$, then $F \leq e^{-2M(p-\theta)^2}$ and so $\mathbb{P}(\text{risky selected}) \leq N_r e^{-2M(p-\theta)^2}$; thus $M \geq \ln(N_r/\delta)/(2(p-\theta)^2)$ particles suffice for error probability $\leq \delta$: the required M grows only logarithmically in N_r .

Proof. (a) The risky candidate i beats the safe one iff $c_r + \kappa K_i/M \leq c_s$, i.e. $K_i \leq M\theta$, i.e. $K_i \leq k^*$ (integer K_i). A risky candidate is selected iff at least one risky candidate does so, and the K_i are independent: $\mathbb{P} = 1 - \mathbb{P}(\forall i : K_i > k^*) = 1 - (1 - F)^{N_r}$. (b) $x \mapsto 1 - (1 - F)^x$ is nondecreasing and strictly increasing if $F \in (0, 1)$; $F \geq \mathbb{P}(K = 0) = (1 - p)^M$ with equality iff $k^* = 0$; $F < 1$ iff $k^* < M$ iff $\theta < 1$. (c) $(1 - F)^{N_r} = e^{-\lambda N_r}$ and $-\ln(1 - F) \geq F$; solve $e^{-\lambda N_r} \leq \delta$. (d) Since $k^*/M \leq \theta < p$, Hoeffding's inequality [5] gives $\mathbb{P}(K \leq k^*) = \mathbb{P}(K/M - p \leq -(p - k^*/M)) \leq e^{-2M(p - k^*/M)^2} \leq e^{-2M(p - \theta)^2}$; then $1 - (1 - F)^{N_r} \leq N_r F$. $\square$

Theorem 11 (Excess cost and optimism; New). In the *mistake regime* $p > \theta$ the best action is the safe one (true cost c_s), and the planner's selected plan has true cost $c_s + \kappa(p - \theta) \mathbf{1}\{\text{risky selected}\}$. Hence

$$\mathbb{E}[\text{true cost of selected plan}] - c_s = \kappa(p - \theta)[1 - (1 - F)^{N_r}] \uparrow \kappa(p - \theta) = J_r - c_s \quad (N_r \rightarrow \infty),$$

$$\mathbb{E}[\min_i \hat{J}_i] = c_r + \frac{\kappa}{M} \sum_{k=1}^M (1 - F_{\text{Bin}}(k - 1; M, p))^{N_r} \downarrow c_r \quad (N_r \rightarrow \infty).$$

So in the limit the planner is wrong with probability one, loses the full gap $J_r - c_s$ and believes that the plan costs c_r , an optimism of κp (the winner's curse or optimiser's curse [13]).

Proof. All risky candidates have the same true cost J_r , so the true cost of the selected plan is J_r if a risky plan is selected and c_s otherwise; use Theorem 10(a). For the second line, $\mathbb{E} \min_i K_i = \sum_{k=1}^M \mathbb{P}(\min_i K_i \geq k) = \sum_{k=1}^M (1 - F_{\text{Bin}}(k - 1; M, p))^{N_r}$, each term is in $[0, 1]$ and tends to 0 because $\mathbb{P}(K \leq k - 1) \geq \mathbb{P}(K = 0) > 0$ for $k \geq 1$. $\square$

Remark 12. If the risky candidates have different failure probabilities p_i , the same proof gives $\mathbb{P}(\text{risky selected}) = 1 - \prod_i (1 - F_{\text{Bin}}(k^*; M, p_i))$. The mechanism needs only (i) independent estimation errors and (ii) selection of an extremum; it is not specific to Bernoulli penalties. With common random numbers across candidates the independence in (a) fails and the formula is no longer exact.

Theorem 13 (Fresh-sample verification of the top- K ; New). *Run stage 1 as above with M particles per risky candidate and keep the set S of the K risky candidates with the smallest $\hat{J}_i$ (ties broken uniformly at random, $N_r \geq K$). Stage 2 draws M' fresh independent particles for each $i \in S$, giving estimates $\hat{J}'_i$, and the final choice is the minimiser of $\{\hat{J}'_i : i \in S\} \cup \{c_s\}$ (ties to risky). Let $F' = F_{\text{Bin}}(\lfloor M'\theta \rfloor; M', p)$.*

- (a) (unbiasedness) *Conditionally on the stage-1 data, $\mathbb{E}[\hat{J}'_i \mid \text{stage 1}] = J_r$ for every $i \in S$; the stage-1 estimate of a selected candidate is biased (Theorem 11).*
- (b) (risk) $\mathbb{P}(\text{risky selected}) = 1 - (1 - F')^K$, which does not depend on N_r or on M . The excess true cost is $\kappa(p - \theta)[1 - (1 - F')^K] \leq \kappa(p - \theta) K e^{-2M'(p-\theta)^2}$ (for $p > \theta$), so $M' \geq \ln(K/\delta)/(2(p - \theta)^2)$ gives risk $\leq \delta$.
- (c) (residual optimism) $\mathbb{E}[\min_{i \in S} \hat{J}'_i] = c_r + \frac{\kappa}{M'} \sum_{k=1}^{M'} (1 - F_{\text{Bin}}(k - 1; M', p))^K$; the bias is exactly 0 for $K = 1$ and is the same function as in Theorem 11 evaluated at (K, M') instead of (N_r, M) .

Proof. S is a function of the stage-1 data and of independent tie-breaking variables, and the stage-2 particles are independent of both. Hence, conditionally on stage 1, the stage-2 failure counts of the members of S are i.i.d. $\text{Bin}(M', p)$, which gives (a) and the conditional law needed for (b): the final choice is risky iff $\min_{i \in S} K'_i \leq \lfloor M'\theta \rfloor$, with probability $1 - (1 - F')^K$ whatever S is. The bound follows as in Theorem 10(d); (c) is the computation of Theorem 11 with $(N_r, M) \rightarrow (K, M')$. $\square$

Remark 14 (The safe fallback must stay in the final comparison). *If the safe plan is only one of the $N_r + 1$ candidates ranked in stage 1 and the final choice is taken among the K best without re-inserting it, verification does not protect. Let $R_1 \sim \text{Bin}(N_r, F)$ be the number of risky candidates ranked ahead of the safe one (ties to risky) and $N_r \geq K$. Then*

$$\mathbb{P}(\text{risky selected}) = \mathbb{P}(R_1 \geq K) + \mathbb{P}(R_1 < K)[1 - (1 - F')^{K-1}] \xrightarrow{N_r \rightarrow \infty} 1,$$

because for $R_1 \geq K$ the set S contains only risky plans and one of them is chosen regardless of the fresh estimates. (Proof: condition on R_1 ; if $R_1 < K$ the verified set contains the safe plan and exactly $K - 1$ risky ones.) Verification is therefore only useful relative to an independently evaluated alternative (an incumbent, a fallback, or an absolute threshold).

Proposition 15 (Monotone shrinkage and ranking invariance; New). *Let $\hat{J}_1, \dots, \hat{J}_N$ be estimates based on $M_1, \dots, M_N$ particles and let $\tilde{J}_i = \Phi(\hat{J}_i; M_i, \hat{\eta})$ where $\hat{\eta}$ is any (data-dependent) hyper-parameter common to all candidates and $x \mapsto \Phi(x; M, \eta)$ is strictly increasing.*

- (a) *If $M_1 = \dots = M_N$, then $\tilde{J}_i < \tilde{J}_j \iff \hat{J}_i < \hat{J}_j$ for all i, j : shrinkage cannot change the arg min, nor any ranking (so it cannot change $\mathbb{P}(\text{risky selected})$ in Theorem 10). This covers the normal-normal empirical-Bayes rule $\tilde{J}_i = \hat{m}_0 + B(\hat{\tau}^2)(\hat{J}_i - \hat{m}_0)$, $B = \hat{\tau}^2/(\hat{\tau}^2 + \sigma^2/M) \in [0, 1]$ with $\hat{m}_0, \hat{\tau}^2$ estimated from the pooled data, and James–Stein-type rules [2] (for $B > 0$; $B = 0$ makes all candidates tie).*
- (b) *If the M_i differ (e.g. after racing), the arg min can change. Example: $\sigma^2 = 4$, $\tau^2 = 1$, $m_0 = 2$, so $B(M) = \tau^2/(\tau^2 + \sigma^2/M)$; candidate 1 has $\hat{J}_1 = 0.9$ from $M_1 = 4$, candidate 2 has $\hat{J}_2 = 1.0$ from $M_2 = 64$. Raw: candidate 1; shrunk: $\tilde{J}_1 = 2 + \frac{1}{2}(0.9 - 2) = 1.45 > \tilde{J}_2 = 2 + \frac{16}{17}(1 - 2) = 1.0588$: candidate 2.*
- (c) *Equal M is not sufficient if the shrinkage factor depends on the candidate (e.g. on per-candidate variance estimates): the map is then not common.*

Proof. (a) With equal M every candidate is mapped by the same strictly increasing function $\Phi(\cdot; M, \hat{\eta})$. (b) is arithmetic: $B(4) = 1/2$, $B(64) = 16/17$. (c) is exhibited numerically in Section 1.7. $\square$

Remark 16. *Racing and shrinkage thus can alter the ranking only through the unequal-sample-size channel. They do not remove the extremum-selection bias of Theorem 11, which acts on the order statistic of the raw estimates; unbiased protection requires fresh data (Theorem 13). Whether unequal- M shrinkage helps or hurts depends on the allocation rule; we make no claim of improvement. Pooling stage-1 and stage-2 samples into one estimate (as in a racing scheme that re-uses samples) does not restore unbiasedness, because the stage-1 part is selected on.*

1.4 T3: energy-score training, identifiability and consistency

For P on $\mathbb{R}^d$ with $\mathbb{E}_P\|X\| < \infty$ and $y \in \mathbb{R}^d$ the *energy score* (with exponent 1) is $\text{ES}(P, y) = \mathbb{E}_P\|X - y\| - \frac{1}{2}\mathbb{E}_P\|X - X'\|$, $X, X' \sim P$ independent; for a point mass $\text{ES}(\delta_m, y) = \|m - y\|$. The *energy distance* is $\mathcal{E}(P, Q) = 2\mathbb{E}\|X - Y\| - \mathbb{E}\|X - X'\| - \mathbb{E}\|Y - Y'\|$ ($X, X' \sim P, Y, Y' \sim Q$, all independent).

Theorem 17 (Strict propriety of the energy score; Known [4, 14, 15]). *For P, Q with finite first moments and $Y \sim Q$,*

$$\mathbb{E} \text{ES}(P, Y) - \mathbb{E} \text{ES}(Q, Y) = \frac{1}{2} \mathcal{E}(P, Q), \quad \mathcal{E}(P, Q) = \frac{1}{c_d} \int_{\mathbb{R}^d} \frac{|\phi_P(t) - \phi_Q(t)|^2}{\|t\|^{d+1}} dt \geq 0,$$

with $c_d = \pi^{(d+1)/2}/\Gamma((d+1)/2)$ and ϕ the characteristic function. Hence $\mathcal{E}(P, Q) = 0 \iff P = Q$, i.e. ES is strictly proper relative to the class of laws with finite first moment.

Proof (recalled for completeness). The first identity is algebra: $\mathbb{E} \text{ES}(Q, Y) = \mathbb{E}\|Y - Y'\| - \frac{1}{2}\mathbb{E}\|Y - Y'\| = \frac{1}{2}\mathbb{E}\|Y - Y'\|$, and $\mathbb{E} \text{ES}(P, Y) = \mathbb{E}\|X - Y\| - \frac{1}{2}\mathbb{E}\|X - X'\|$. For the second use $\|x\| = c_d^{-1} \int (1 - \cos\langle t, x \rangle) \|t\|^{-d-1} dt$ [15] and $\mathbb{E} \cos\langle t, X - Y \rangle = \text{Re } \phi_P(t) \overline{\phi_Q(t)}$, $\mathbb{E} \cos\langle t, X - X' \rangle = |\phi_P(t)|^2$; by Fubini (finite first moments) the integrand becomes $-2 \text{Re } \phi_P \phi_Q + |\phi_P|^2 + |\phi_Q|^2 = |\phi_P - \phi_Q|^2$. A law is determined by its characteristic function. $\square$

Noise-input models. Let $Q_\theta(\cdot | x)$ be the law of $f_\theta(x, U)$, $U \sim \gamma = \mathcal{N}(0, I_k)$, and let data $(X, Y) \sim P_{XY}$. Define the population risk $L(\theta) = \mathbb{E} \text{ES}(Q_\theta(\cdot | X), Y)$ and $L_\star = \frac{1}{2}\mathbb{E}\|Y - Y'\|$ (Y' an independent draw from $P_{Y|X}$ given X). By Theorem 17 applied conditionally on X ,

$$L(\theta) - L_\star = \frac{1}{2} \mathbb{E}_X \mathcal{E}(Q_\theta(\cdot | X), P_{Y|X}) \geq 0, \quad (1)$$

with equality iff $Q_\theta(\cdot | X) = P_{Y|X}$ a.s.

Proposition 18 (The law is identified, the noise map is not; New). *(a) If $T : \mathbb{R}^k \rightarrow \mathbb{R}^k$ is measurable with $T_\# \gamma = \gamma$ (any orthogonal map, a sign flip, or an x -dependent family T_x), then $f_\theta(x, T_x(\cdot))$ induces the same Q_θ and the same L . The minimisers of L are therefore never isolated points in θ -space when the class is closed under such T : the identified object is $\theta \mapsto Q_\theta$. (b) If $k \geq d$, every law P on $\mathbb{R}^d$ is $f_\# \gamma$ for some measurable f (Rosenblatt transformation [12]; for $k < d$ a Borel isomorphism argument, cf. [8]), so no structure of f beyond its pushforward is determined. (c) Everything that depends on candidates only through their marginal laws is identified: $J_\kappa(a)$, p_a , $\text{CVaR}_\alpha(c_\kappa(Z_a))$ for one-step models, and, for rollouts with independent noise at each step, the law of Z_H (it is determined by the one-step kernels). The coupling induced by sharing U between candidates (common random numbers) is not identified: let $Z_1 = \mu_1 + sU$, $Z_2 = \mu_2 + \epsilon sU$ with $\epsilon \in \{+1, -1\}$ and $U \sim \mathcal{N}(0, 1)$; the marginal laws do not depend on ϵ but, with $m_i = \mu_i - g$,*

$$\text{Var}(c(Z_1) - c(Z_2)) = 4s^2 (m_1 - \epsilon m_2)^2.$$

For $m_1 m_2 > 0$ the variance under $\epsilon = +1$ is smaller than under $\epsilon = -1$ by the factor $((m_1 - m_2)/(m_1 + m_2))^2$: the variance reduction promised by common random numbers is a property of the parametrisation, not of the learned law.

Proof. (a) $\mathbb{E} \text{ES}(Q, Y)$ depends on f_θ only through Q_θ . (b) Rosenblatt: composing the inverse conditional distribution functions with the coordinates $\Phi(u_j)$ of u . (c) $c(Z_1) - c(Z_2) = m_1^2 - m_2^2 + 2sU(m_1 - \epsilon m_2) + s^2(1 - \epsilon^2)U^2$ and $\epsilon^2 = 1$, so the variance is $4s^2(m_1 - \epsilon m_2)^2$. $\square$

Theorem 19 (Consistency of the empirical energy-score minimiser; New, standard M-estimation argument [7, 9]). *Let $(X_i, Y_i)_{i \leq n}$ be i.i.d. from P_{XY} and, independently, let U_{ij} ($j \leq m$, $m \geq 2$) be i.i.d. γ . Let*

$$\hat{L}_n(\theta) = \frac{1}{n} \sum_{i=1}^n \left[\frac{1}{m} \sum_j \|f_\theta(X_i, U_{ij}) - Y_i\| - \frac{1}{2m(m-1)} \sum_{j \neq l} \|f_\theta(X_i, U_{ij}) - f_\theta(X_i, U_{il})\| \right].$$

Assume (A1) Θ is a compact metric space; (A2) $\theta \mapsto f_\theta(x, u)$ is continuous for every (x, u) and $(x, u) \mapsto f_\theta(x, u)$ is measurable; (A3) there is G with $\|f_\theta(x, u)\| \leq G(x, u)$ for all θ , $\mathbb{E}G(X, U) < \infty$ and $\mathbb{E}\|Y\| < \infty$. Let $\Theta_\star = \arg \min_\Theta L$ and let $\hat{\theta}_n$ be any measurable ε_n -minimiser of $\hat{L}_n$ with $\varepsilon_n \rightarrow 0$.

(a) *L is continuous, $\Theta_\star \neq \emptyset$ is compact, and $\text{dist}(\hat{\theta}_n, \Theta_\star) \rightarrow 0$ a.s.*

(b) *(well-specified case, A4: some θ_0 has $Q_{\theta_0}(\cdot | X) = P_{Y|X}$ a.s.) $\Theta_\star = \{\theta : Q_\theta(\cdot | X) = P_{Y|X} \text{ a.s.}\}$, and the learned predictive laws converge: $\mathbb{E}_X \mathcal{E}(Q_{\hat{\theta}_n}(\cdot | X), P_{Y|X}) \rightarrow 0$ a.s.*

In general $\Theta_\star$ is a set (Proposition 18); $\hat{\theta}_n$ itself converges only when a normalisation makes the minimiser unique.

Proof. Envelope. The second term of the summand is at most $\frac{1}{m} \sum_j G(X_i, U_{ij})$ (since $\sum_{j \neq l} (G_j + G_l) = 2(m-1) \sum_j G_j$), so $|\ell_\theta| \leq D := \frac{2}{m} \sum_j G(X_i, U_{ij}) + \|Y_i\|$ with $\mathbb{E}D = 2\mathbb{E}G + \mathbb{E}\|Y\| < \infty$. *Unbiasedness.* Since the U_{ij} are i.i.d., $\mathbb{E}[\ell_\theta | X_i, Y_i] = \mathbb{E}\|X - y\| - \frac{1}{2}\mathbb{E}\|X - X'\| = \text{ES}(Q_\theta(\cdot | X_i), Y_i)$, so $\mathbb{E}\hat{L}_n(\theta) = L(\theta)$. *Uniform LLN.* By (A1)–(A3), dominated convergence gives continuity of L , and the uniform law of large numbers for continuous dominated summands [7, 9] gives $\sup_\theta |\hat{L}_n - L| \rightarrow 0$ a.s. *Argmin.* $\Theta_\star$ is nonempty and compact by continuity. For $\delta > 0$ let $\Theta_\delta = \{\theta : \text{dist}(\theta, \Theta_\star) \geq \delta\}$ (compact); if nonempty, $\eta := \min_{\Theta_\delta} L - \min_\Theta L > 0$. Choose n so large that $\sup |\hat{L}_n - L| < \eta/3$ and $\varepsilon_n < \eta/3$. For $\theta \in \Theta_\delta$, $\hat{L}_n(\theta) > \min L + 2\eta/3$, whereas $\hat{L}_n(\hat{\theta}_n) \leq \min \hat{L}_n + \varepsilon_n < \min L + 2\eta/3$, so $\hat{\theta}_n \notin \Theta_\delta$. (b) By (1) and A4, $\min L = L_\star$ and $L(\theta) = L_\star$ iff $\mathbb{E}_X \mathcal{E} = 0$ iff the laws agree a.s. (as $\mathcal{E} \geq 0$). Moreover $\mathbb{E}_X \mathcal{E}(Q_\theta, P) = 2(L(\theta) - L_\star)$ is continuous in θ and vanishes on $\Theta_\star$, so it tends to 0 along $\hat{\theta}_n$. $\square$

Remark 20 (Which energy score? Practical caveat; New). *The summand above uses the unbiased pair average (as in our training code). The common “V-statistic” variant, with the double sum over all (j, l) including $j = l$, multiplies the spread term by $\rho_m = (m-1)/m$ and is not proper [3]. For the Gaussian scale family with truth $\mathcal{N}(0, 1)$ its population risk is $R_V(s) = \sqrt{2/\pi} \sqrt{1+s^2} - \rho_m s / \sqrt{\pi}$, minimised at $s_m^2 = \rho_m^2 / (2 - \rho_m^2)$: $s_2 = 0.378$, $s_4 = 0.626$, $s_8 = 0.788$, $s_{32} = 0.940$.*

Remark 21 (Point-mass predictors). *Training a deterministic predictor with the energy score gives $\text{ES}(\delta_m, y) = \|m - y\|$, i.e. the spatial median, whereas MSE gives the mean; for skewed noise the two differ (for $\text{Exp}(1)$ noise: mean 1, median $\ln 2$). Neither is the right input for expected-cost planning unless the cost is linear.*

1.5 T4: open-loop versus closed-loop planning under process noise

Model. $x_{t+1} = x_t + u_t + \sigma w_t$, $t = 0, \dots, H-1$, w_t i.i.d. $\mathcal{N}(0, 1)$, x_0 given, goal g . An *open-loop* (OL) policy is a deterministic sequence (u_t) (this is what a sampling planner evaluates when it rolls candidate action sequences through a noisy model); a *closed-loop* (CL) policy has $u_t = \pi_t(x_0, \dots, x_t)$. For any criterion $V_{\text{OL}}(\rho) = \inf_{\text{OL}} \rho(C) \geq V_{\text{CL}}(\rho) = \inf_{\text{CL}} \rho(C)$ since OL policies are CL policies.

Proposition 22 (Variance growth; New). (a) For every OL plan $\text{Var}(x_t) = t\sigma^2$. (b) For the feedback $u_t = \bar{u}_t - k(x_t - \bar{x}_t)$ with constant gain $k \in (0, 2)$ and nominal path $\bar{x}$, the deviation $e_t = x_t - \bar{x}_t$ satisfies $e_{t+1} = (1 - k)e_t + \sigma w_t$, so $\text{Var}(x_t) = \sigma^2 \frac{1 - (1-k)^{2t}}{1 - (1-k)^2} \leq \frac{\sigma^2}{k(2-k)}$, and for $k = 1$ (deadbeat) $\text{Var}(x_t) = \sigma^2$. (c) For a threshold $d = B - g > 0$ and the policies “hold at g ”: $\mathbb{P}(x_t \geq B) = \bar{\Phi}(d/(\sigma\sqrt{t})) \rightarrow \frac{1}{2}$ for OL, whereas for the feedback in (b) $\mathbb{P}(x_t \geq B) \leq \bar{\Phi}(d\sqrt{k(2-k)}/\sigma)$ for all t .

Proof. (a) $x_t = \bar{x}_t + \sigma \sum_{s < t} w_s$. (b) $e_t = \sigma \sum_{s < t} (1 - k)^{t-1-s} w_s$ is centred Gaussian with the stated variance (geometric sum). (c) Gaussian tails; the second variance is increasing in t and bounded by its limit. $\square$

Definition 23. A functional ρ of nonnegative random variables is FSD-monotone if $Y_1 \leq_{\text{st}} Y_2 \Rightarrow \rho(Y_1) \leq \rho(Y_2)$, and positively homogeneous if $\rho(\lambda Y) = \lambda \rho(Y)$ for $\lambda > 0$. Examples: $\mathbb{E}$, VaR_α , CVaR_α and all spectral risk measures.

Theorem 24 (LQG: open loop costs a factor H ; New). Let $C = (x_H - g)^2$ and let ρ be FSD-monotone and positively homogeneous. Then

$$V_{\text{OL}}(\rho) = H\sigma^2 \rho(\chi_1^2), \quad V_{\text{CL}}(\rho) = \sigma^2 \rho(\chi_1^2), \quad V_{\text{OL}}/V_{\text{CL}} = H,$$

attained by any OL plan with $\bar{x}_H = g$ and by any CL policy with $u_{H-1} = g - x_{H-1}$, respectively. For $\rho = \mathbb{E}$: $V_{\text{OL}} = H\sigma^2$, $V_{\text{CL}} = \sigma^2$; for $\rho = \text{CVaR}_\alpha$: $V_{\text{CL}} = \sigma^2 c_\alpha$ with c_α from Corollary 8. In particular, if $0 < \rho(\chi_1^2) < \infty$ then for every $H \geq 2$ the optimal open-loop plan is strictly worse than the optimal closed-loop policy under every such criterion, including every CVaR_α , and a planner that scores candidates by open-loop rollouts but is executed with replanning overestimates the achievable cost by the factor H .

Proof. Lower bound, CL. For any non-anticipative policy, $x_H - g = c + \sigma w_{H-1}$ with $c = x_{H-1} + u_{H-1} - g$ measurable with respect to $\mathcal{F}_{H-1}$ and w_{H-1} independent of $\mathcal{F}_{H-1}$. For fixed real c and $r \geq 0$, $\mathbb{P}(|c + \sigma w| \leq r) = \Phi((r - c)/\sigma) - \Phi((-r - c)/\sigma)$ is maximal at $c = 0$ (the integral of a symmetric unimodal density over a symmetric interval shifted by c is maximal at $c = 0$; [1]). Hence $\mathbb{P}(C \leq r^2 \mid \mathcal{F}_{H-1}) \leq \mathbb{P}(\sigma^2 w^2 \leq r^2)$ and, integrating, $C \geq_{\text{st}} \sigma^2 \chi_1^2$; FSD-monotonicity and homogeneity give $\rho(C) \geq \sigma^2 \rho(\chi_1^2)$, with equality for $u_{H-1} = g - x_{H-1}$ (then $C = \sigma^2 w_{H-1}^2$). OL. For a deterministic plan $x_H - g \sim \mathcal{N}(m, H\sigma^2)$ with $m = \bar{x}_H - g$; the same argument with $c = m$ and standard deviation $\sigma\sqrt{H}$ gives $C \geq_{\text{st}} H\sigma^2 \chi_1^2$, attained at $m = 0$. $\square$

Remark 25 (Entropic risk). For $\rho_\vartheta(C) = \vartheta^{-1} \ln \mathbb{E} e^{\vartheta C}$ (FSD-monotone, not homogeneous; cf. [6, 16]) the same argument gives $V_{\text{CL}} = -\frac{1}{2\vartheta} \ln(1 - 2\vartheta\sigma^2)$ and $V_{\text{OL}} = -\frac{1}{2\vartheta} \ln(1 - 2\vartheta H\sigma^2)$, the latter being $+\infty$ for $\vartheta \geq 1/(2H\sigma^2)$: open-loop planning loses finiteness of the risk-sensitive criterion H times earlier.

Absorbing boundary. Now let the walk be absorbed (stopped) at the first time $\tau \in \{1, \dots, H\}$ with $x_\tau \geq B$ (where $g < B$), controls $|u_t| \leq U$, and

$$C = (x_H - g)^2 + \kappa \mathbf{1}\{\tau \leq H\}, \quad x_H = B \text{ if absorbed.}$$

Define the last-step function $q(y) = \mathbb{E}[(\min(y + \sigma w, B) - g)^2 + \kappa \mathbf{1}\{y + \sigma w \geq B\}]$, the expected cost of landing at mean y from an alive state, and the cost of being absorbed earlier, $C_k = (B - g)^2 + \kappa$.

Theorem 26 (Strict open-loop gap with an absorbing boundary; New). Let $\rho = \mathbb{E}$ (expected penalised cost). Then $V_{\text{CL}} \leq V_{\text{OL}}$ with equality for $H = 1$. Assume $\sigma > 0$, $U > 0$, that $\inf q < C_k$ (deliberately failing is not optimal; then q attains its minimum q^* at some y^* , because $q(y) \rightarrow \infty$ as $y \rightarrow -\infty$ and $q(y) \rightarrow C_k$ as $y \rightarrow +\infty$) and that $y^* - U < B$ (“the best landing point is reachable from the alive region”). Then $V_{\text{CL}} < V_{\text{OL}}$ strictly for every $H \geq 2$.

Proof. OL plans form a compact set $[-U, U]^H$ and the OL cost is continuous in the plan, so an optimal OL plan u° exists. Let $y = x_{H-1}$ denote the (alive part of the) state before the last step; by dynamic programming, given $x_{H-1} < B$ and control u the expected cost-to-go is $q(x_{H-1} + u)$. Define the CL policy that follows u° until time $H-2$ and chooses $u_{H-1}(x) \in \arg \min_{|u| \leq U} q(x+u)$ (smallest minimiser, measurable) when alive. Its cost differs from that of u° only through

$$\text{gap} = \mathbb{E}[(q(x_{H-1} + u_{H-1}^\circ) - \underline{q}(x_{H-1})) \mathbf{1}\{\text{alive}\}] \geq 0, \quad \underline{q}(x) = \min_{|u| \leq U} q(x+u).$$

For $H \geq 2$ the alive part of x_{H-1} has a density that is strictly positive on $(-\infty, B)$ (a killed Gaussian walk: each step has a strictly positive Gaussian kernel). If $\text{gap} = 0$, then $q(x + u_{H-1}^\circ) = \underline{q}(x)$ for almost every $x < B$. Take the nonempty open interval $I = \{x < B : |x - y^*| < U\}$ (nonempty because $y^* - U < B$); for $x \in I$ the window $[x - U, x + U]$ contains y^* , hence $\underline{q}(x) = q^*$, so $q = q^*$ on a nonempty open interval (by continuity of q). But q is real-analytic ($q(y) = \int h(z) \sigma^{-1} \varphi((z - y)/\sigma) dz$ with h of quadratic growth is a Gaussian convolution) and not constant ($q(y) \rightarrow \infty$ as $y \rightarrow -\infty$), contradicting the identity theorem. So $\text{gap} > 0$ and $V_{\text{CL}} \leq \text{cost}(\text{modified policy}) = V_{\text{OL}} - \text{gap} < V_{\text{OL}}$. For $H = 1$ the only decision is taken at x_0 and $\text{OL} = \text{CL}$. $\square$

Remark 27 (Scope, and what is *not* true). (i) *Theorem 26 is proved for the expected penalised cost; the failure penalty κ is itself a form of risk aversion, and Theorem 24 covers CVaR without a barrier. For CVaR with a barrier closed-loop optimisation is time-inconsistent and we make no claim.* (ii) *Feedback bounds the spread (Proposition 22) and the time- t exceedance probability, but not the cumulative first-passage probability: for any stationary Gaussian closed loop the probability of ever reaching B tends to 1 as the horizon grows. In our Monte-Carlo check, holding at g with the weak gain $k = 0.5$ has first-passage probability below that of open loop up to $t = 20$, but above it at $t = 40$ (Section 1.7). Our first conjecture, that feedback keeps the cumulative failure probability bounded, was therefore false and is not claimed.* (iii) *In the LQG model without barrier all candidates have the same open-loop variance $H\sigma^2$, so Theorem 5 predicts no mean-embedding regret there; the open-loop penalty is a separate effect (value of feedback), which in the barrier model interacts with the variance-dependent failure probability.*

1.6 T5: scale recalibration cannot repair a change of shape

Let P be the model's predictive law of a scalar latent coordinate and Q the true law after a distribution shift; failure is $F_B = \{z \geq B\}$ (thresholds B differ across candidates, since candidates pass the hazard at different distances). A *scale recalibration* replaces P by P_s , the law of $\mu_P + s(Z - \mu_P)$, $Z \sim P$, $s > 0$ (location-scale recalibration also frees the location).

Theorem 28 (Scale recalibration and failure events; New). *Let $P = \mathcal{N}(0, 1)$.*

- (a) *(moment blindness) If Q has mean 0 and variance 1, every recalibration rule that uses Q only through its first two moments (variance matching, Gaussian likelihood or second-moment calibration, per-dimension or full covariance matching in higher dimension) returns $s = 1$, $P_s = P$, and $P(F_B) \neq Q(F_B)$ for any B at which the two laws differ.*
- (b) *(tail mismatch, one-sided) Let Q be the Laplace law with variance 1, $\bar{\Phi}_Q(B) = Q(Z \geq B) = \frac{1}{2}e^{-\sqrt{2}B}$ for $B \geq 0$. For every $s > 0$, $P_s(F_B)/Q(F_B) \leq e^{-B^2/(2s^2) + \sqrt{2}B} \rightarrow 0$ as $B \rightarrow \infty$. The scale that matches $Q(F_B)$ exactly, $s_B = B/\bar{\Phi}^{-1}(\bar{\Phi}_Q(B))$, is not constant: $s_B^2 \geq B/(2\sqrt{2}) \rightarrow \infty$, while $s_B < 1$ for small B .*
- (c) *(decision conflict, general form) Suppose $B \mapsto s_B$ is not constant. Then there are two thresholds $B_1, B_2 > 0$ and levels θ_1, θ_2 such that the decisions “accept the candidate with hazard at B_j iff its predicted failure probability is below θ_j ” ($j = 1, 2$) are both correct under Q for no scale s . Concretely for $B_1 = 1$, $\theta_1 = 0.14$, $B_2 = 3$, $\theta_2 = 0.004$: the truth is accept*

at B_1 ($Q(F_1) = 0.1216 < 0.14$) and reject at B_2 ($Q(F_3) = 0.00718 > 0.004$); the Gaussian with scale s accepts at B_1 iff $s < s_a = 0.9257$ and rejects at B_2 iff $s > s_b = 1.1312$, and $s_b > s_a$.

- (d) (error floor, two-point law) Let $Q = \frac{1}{2}(\delta_{-1} + \delta_1)$ (mean 0, variance 1, the shape of the benchmark's bimodal wind). For every $s > 0$, $\sup_x |F_Q(x) - \Phi(x/s)| \geq \frac{1}{4}$; in particular $\sup_B |P_s(F_B) - Q(F_B)| \geq \frac{1}{4}$ for all scales. (For the unit-variance Laplace law the floor $\inf_s \sup_x |F_Q - \Phi(\cdot/s)|$ is 0.028, attained near $s = 0.81$.)

The failure probability is a functional of the whole shape of the law near the hazard; scale (or any affine) recalibration fixes a two-parameter summary and leaves the shape error untouched.

Proof. (a) By construction. (b) $\bar{\Phi}(x) \leq \frac{1}{2}e^{-x^2/2}$ for $x \geq 0$, so $P_s(F_B) = \bar{\Phi}(B/s) \leq \frac{1}{2}e^{-B^2/(2s^2)}$, and divide by $\frac{1}{2}e^{-\sqrt{2}B}$. Applying the same bound at $s = s_B$ to $\bar{\Phi}(B/s_B) = \frac{1}{2}e^{-\sqrt{2}B}$ gives $\sqrt{2}B \geq B^2/(2s_B^2)$, i.e. $s_B^2 \geq B/(2\sqrt{2})$. Small B : the entries of Table 2. (c) Pick B_1, B_2 with $s_{B_1} < s_{B_2}$ (possible because s_B is not constant; the thresholds need not be ordered). $s \mapsto P_s(F_B) = \bar{\Phi}(B/s)$ is continuous and strictly increasing, so $P_s(F_{B_j}) < \theta$ iff $s < s(\theta; B_j) := B_j/\bar{\Phi}^{-1}(\theta)$. Choose θ_1 slightly above $Q(F_{B_1})$ (truth: accept) and θ_2 slightly below $Q(F_{B_2})$ (truth: reject). Correctness at B_1 requires $s < s(\theta_1; B_1)$, which is slightly above s_{B_1} ; correctness at B_2 requires $s > s(\theta_2; B_2)$, slightly below s_{B_2} . For small enough offsets $s(\theta_1; B_1) < s(\theta_2; B_2)$, so no s satisfies both. The numbers are the evaluation of the two thresholds. (d) F_Q jumps from $\frac{1}{2}$ to 1 at $x = 1$ while $x \mapsto \Phi(x/s)$ is continuous; at $x = 1$ the values $F_Q(1^-) = \frac{1}{2}$ and $F_Q(1) = 1$ cannot both be within $\frac{1}{4}$ of $\Phi(1/s)$ unless $\Phi(1/s) = \frac{3}{4}$, in which case the distance is exactly $\frac{1}{4}$. The statement about failure probabilities follows since $Q(F_B) = \frac{1}{2}\mathbf{1}\{B \leq 1\}$ for $B > -1$ and $\sup_B$ includes $B = 1$ and $B = 1^+$. $\square$

B	$Q(Z \geq B)$	$\bar{\Phi}(B) (s = 1)$	ratio	matching s_B
0.5	0.2465	0.3085	1.252	0.730
1.0	0.1216	0.1587	1.305	0.857
2.0	0.0296	0.0228	0.770	1.060
3.0	0.0072	0.00135	0.188	1.226
4.0	0.0018	0.00003	0.018	1.370

Table 2: Gaussian model versus unit-variance Laplace truth: failure probabilities at several thresholds and the scale that would be needed at each (output of `d4_verify.py`, T5).

Remark 29 (Relation to the measurements). *The empirical benchmark finds that an oracle scalar spread rescaling (true factor 1.44) did not beat the unadapted predictor and that per-dimension rescaling did not either. Theorem 28 shows that this is not paradoxical: the shift changes the shape of the predictive law near the hazard (bimodal wind accumulated over the horizon, interacting with the absorbing pit), a scale factor equalises at most one tail level, and the scale that is right for one candidate is wrong for another. This is an explanation consistent with, not a proof of, the observed numbers.*

1.7 Numerical verification

`experiments/d4_verify.py` (single CPU core; output in the directory `results/d4/`: the printed tables in `d4_verify.txt`, a machine-readable summary in `d4_verify.json`) checks every statement above against closed forms, Monte Carlo and, for the barrier model, a grid dynamic program. Table 3 lists the largest error per claim; the Monte-Carlo errors are consistent with sampling noise (two or three standard errors at most).

Three outcomes deserve mention. (1) The conjecture that feedback keeps the cumulative failure probability bounded was contradicted by the first-passage simulation (Remark 27) and

was replaced by the exceedance statement of Proposition 22(c). (2) The first version of the pooled-verification check disagreed with its formula; the discrepancy was a bug in the simulator (a risky plan is selected automatically when the safe plan is not in the verified set), not in Remark 14, and the corrected simulation agrees. (3) Candidate-specific shrinkage with equal M changes the argmin in about 20% of random trials, which is why Proposition 15(a) requires a common map.

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Group	Check	Max. error	Result
T1	bias–variance identity (rel. MC error)	6.1×10^{-4}	pass
T1	MSE fit converges to the mean (sup error, $n = 10^6$)	8.1×10^{-3}	pass
T1	regret identity, Thm 5(a)	4.4×10^{-16}	pass
T1	violations of $R \leq \Delta_v$ (count, 12 instances)	0	pass
T1	MC regret vs exact (abs.)	5.8×10^{-3}	pass
T1	tightness example $R = V - \eta$ (rel.)	1.7×10^{-3}	pass
T1	failure-penalised regret, $\kappa \leq 500$ (rel.)	7.5×10^{-4}	pass
T1	c_α closed form vs MC tail mean (rel.)	1.6×10^{-3}	pass
T1	CVaR regret (rel.)	1.6×10^{-3}	pass
T1	$\mathbb{E}C \leq \text{CVaR}_\alpha \leq \mathbb{E}C/(1 - \alpha)$ (violations)	0	pass
T2	selection probability, 40 grid cells (abs.)	1.3×10^{-3}	pass
T2	monotone in N_r (violations)	0	pass
T2	excess true cost (abs.)	3.0×10^{-3}	pass
T2	$\mathbb{E} \min \hat{J}$ formula (abs.)	8.5×10^{-3}	pass
T2	Hoeffding bound (violations)	0	pass
T2	fresh verification, $1 - (1 - F')^K$ (abs.)	1.8×10^{-3}	pass
T2	pooled variant, Remark 14 (abs.)	9.2×10^{-4}	pass
T2	fresh estimate unbiased given selection (abs.)	3.6×10^{-3}	pass
T2	argmin changed by shrinkage, equal M (fraction)	0	pass
T2	unequal- M flip exhibited	0	pass
T3	ES of a Gaussian, closed form (abs.)	5.6×10^{-4}	pass
T3	ES minimised at truth, (m, s) grid (dist.)	0	pass
T3	$\mathbb{E} \text{ES}(P, Y) - \mathbb{E} \text{ES}(Q, Y) = \frac{1}{2} \mathcal{E}$ (abs.)	1.8×10^{-4}	pass
T3	point-mass ES gives median (abs.)	6.9×10^{-3}	pass
T3	CRN variance $4s^2(m_1 - \epsilon m_2)^2$ (rel.)	2.6×10^{-4}	pass
T3	dist. to $\Theta_\star$ at largest n (median)	2.3×10^{-2}	pass
T3	V-statistic minimiser s_m (abs.)	1.0×10^{-3}	pass
T4	variances in Prop. 22 (rel.)	2.9×10^{-3}	pass
T4	values $H\sigma^2\rho$ vs $\sigma^2\rho$ (rel.)	2.8×10^{-3}	pass
T4	CVaR increasing in $ m $ (violations)	0	pass
T4	entropic risk, $H = 1$ (rel.)	4.8×10^{-4}	pass
T4	barrier DP / OL optimum vs MC (std. errors)	1.9×10^0	pass
T4	min gap $V_{\text{OL}} - V_{\text{CL}}$, $H = 2.5$ (must be > 0)	0.108	pass
T4	hypotheses of Thm 26 hold	0	pass
T4	exceedance probabilities, Prop. 22(c) (abs.)	1.8×10^{-3}	pass
T4	FB first passage $\leq$ OL for $t \leq 20$ (see Remark 27)	0	pass
T5	Laplace tails (rel.)	1.2×10^{-2}	pass
T5	scales fixing both decisions (count, must be 0)	0	pass
T5	floor $\geq 1/4$ (shortfall)	0	pass

Table 3: Largest error per checked claim (`results/d4/d4_verify.json`). Monte-Carlo entries are sampling errors ($S = 2 \times 10^6$ draws unless stated). Tolerances are set at about three standard errors.